CMI Entrance Examination

Mock Test · Part A

Answer all questions

Question 1 *4 points*

Which function(s) among the following can take all complex values, when defined on $\mathbb{C}$?

(a) $f(z) = \sin z$

(b) $f(z) = e^z$

(c) $f(z) = z^2$

(d) $f(z) = ze^z$

Write your answer as a string of letters: like: ABC or AB etc.

Question 2 *4 points*

Write True/False for the following statements

(a) There are no naturals $m, n > 1$ such that $m! = k^n$ for some $k \in \mathbb{N}$.

(b) There are infinitely many triples (a, b, c) of naturals such that $(a!)(b!) = c!$.

(c) There are infinitely many natural numbers n such that the equation $a^b + b^a = n$ has no solutions in natural (a, b).

(d) There are infinitely many n for which $(1! + 2! + \cdots + n!)$ is some perfect power of a positive integer.

Question 3 *4 points*

Write CAN / CANNOT in the blanks

(a) There __________ exist a function $f : \mathbb{R} \to \mathbb{R}$ such that $f(x)+f(y) = x^2+xy+y^2, \forall x, y \in \mathbb{R}$.

(b) There __________ exist a continuous function $f : \mathbb{R} \to \mathbb{R}$ such that $f(f(x)) = -x, \forall x \in \mathbb{R}$.

(c) There __________ exist a function $f : \mathbb{R} \to \mathbb{R}$ such that f is continuous everywhere, but nowhere differentiable.

(d) There __________ exist a function $f : \mathbb{R} \to \mathbb{R}$ which is continuous only at rationals.

Question 4

4 points

Let $P(x)$ be a real polynomial such that $P(x) > 0, \forall x \in \mathbb{R}$. Then

(a) Degree of P is even.

(b) P can be written as a sum of squares of two real polynomials.

(c) The leading coefficient of P is positive.

(d) P has a global minimum.

Write your answer as a string of correct options: like: ACD / BD etc.

Question 5

4 points

Write True/False for the following statements

(a) If $f : \mathbb{R} \to \mathbb{R}$ is a continuous and bounded function, then it must attain a global maximum or a global minimum value.

(b) Every differentiable function $f : \mathbb{R} \to \mathbb{R}$ with a bounded derivative is uniformly continuous on $\mathbb{R}$.

(c) If the improper integral $\int_0^\infty f(x)\,dx$ converges, then it is necessary that $\lim_{x\to\infty} f(x) = 0$.

(d) There exists a real-valued function defined on $\mathbb{R}$ that is differentiable at exactly one point.

Question 6

4 points

Let $S = \{1, 2, 3, \ldots, 10\}$. Which of the following statements are correct?

(a) The number of subsets of S having an even number of elements is exactly 512.

(b) The number of ordered pairs of subsets (A, B) of S such that A is a strictly proper subset of B is $3^{10} - 2^{10}$.

(c) The number of subsets of S that contain no two consecutive integers is 128.

(d) The number of ways to choose two disjoint, non-empty subsets of S is $\frac{3^{10}-2^{11}+1}{2}$.

Write your answer as a string of correct options: like: ACD / BD etc.

Question 7

4 points

Write CAN / CANNOT in the blanks

(a) There __________ exist a real number x satisfying $\sin^4 x + \cos^4 x = \frac{1}{4}$.

(b) There __________ exist a triangle ABC for which $\cos^2 A + \cos^2 B + \cos^2 C < 1$.

(c) There __________ exist a real number x such that $\cos(\cos x) = \sin(\sin x)$.

(d) There __________ exist a non-degenerate triangle ABC where $\sin A + \sin B = \sin C$.

Question 8 *4 points*

Write True/False for the following statements

(a) If all the vertices of a convex pentagon lie on a circle, then at least one of its interior angles is strictly greater than $100°$.

(b) There exists a non-degenerate triangle with side lengths 3, 4, and 8.

(c) The locus of points from which the tangents drawn to a parabola are perpendicular to each other is a straight line.

(d) It is possible to completely tile the Euclidean plane using only congruent regular octagons.

Question 9 *4 points*

Write True/False for the following statements

(a) For all positive real numbers a, b, and c, the inequality $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$ holds.

(b) For all positive real numbers a, b, and c, it is guaranteed that $\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{a} \geq a + b + c$.

(c) If a, b, and c are the side lengths of a non-degenerate triangle, then $a^2+b^2+c^2 < 2(ab+bc+ca)$.

(d) For all positive real numbers a, b, and c satisfying $abc = 1$, it is always true that $a + b + c \geq ab + bc + ca$.

Question 10 *4 points*

Let A and B be two events associated with a random experiment. Which of the following statements are correct?

(a) If $P(A \cup B) = P(A) + P(B)$, then A and B must be independent events.

(b) If A and B are independent and $P(A) \in (0, 1), P(B) \in (0, 1)$, then they are not mutually exclusive.

(c) $P(A \cap B) \geq P(A) + P(B) - 1$.

(d) If $P(A \mid B) > P(A)$, then $P(B \mid A) > P(B)$.

Write your answer as a string of correct options: like: ACD / BD etc.

End of Part A

CMI Entrance Examination

Mock Test · Part B

Answer all questions. Show all your work for full credit.

Question 1

15 points

Let $n \geq 3$ be an integer, and let $f : [0, n] \to \mathbb{R}$ be a continuously differentiable function. Suppose that $f(0) = 0$ and $f(n) = 0$, and that at every interior integer point the function takes one of the two extreme values: $f(k) \in \{-1, 1\}$ for every $k \in \{1, 2, \dots, n-1\}$.

By the Mean Value Theorem, for each $k \in \{1, 2, \dots, n\}$ there exists a point $c_k \in (k-1, k)$ such that

$$f'(c_k) \;=\; f(k) - f(k-1).$$

Define the *derivative energy* sum

$$E \;=\; \sum_{k=1}^{n} \bigl[f'(c_k)\bigr]^2.$$

(a) Determine all possible numerical values of the term $[f'(c_k)]^2$ in the three cases $k = 1$, $k = n$, and $2 \leq k \leq n-1$.

(b) Using your observations from part (a), prove that $E \equiv 2 \pmod 4$, regardless of the specific sequence of values in $\{-1, 1\}$ chosen at the interior points.

Question 2

15 points

Let $f : [0, 1] \to \mathbb{R}$ be a continuously differentiable function.

(a) Prove that

$$\lim_{n\to\infty} n \int_0^1 x^n f(x)\, dx \;=\; f(1).$$

(b) Determine, with proof, the value of the following limit in terms of $f(1)$ and $f'(1)$:

$$\lim_{n\to\infty} n \left(n \int_0^1 x^n f(x)\, dx \;-\; f(1) \right).$$

Question 3

15 points

Let p be a prime number with $p \equiv 1 \pmod 4$.

(a) Prove that there exists an integer x such that $x^2 \equiv -1 \pmod p$.

(b) Let $N = \lfloor \sqrt{p} \rfloor$. Using the Pigeonhole Principle, prove that there exist two distinct pairs of integers (u_1, v_1) and (u_2, v_2) with $0 \le u_i, v_i \le N$ such that

$$u_1 - v_1 x \equiv u_2 - v_2 x \pmod{p}.$$

(c) Let $a = |u_1 - u_2|$ and $b = |v_1 - v_2|$. Prove that $a^2 + b^2 = p$.

Question 4

15 points

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a continuous function satisfying the parallelogram identity

$$f(u+v) + f(u-v) = 2f(u) + 2f(v)$$

for all $u, v \in \mathbb{R}^2$. Suppose further that there exists a continuous function $g : [0, \infty) \to \mathbb{R}$ such that

$$f(v) = g(\|v\|) \qquad \text{for every } v \in \mathbb{R}^2,$$

where $\|v\| = \sqrt{v_1^2 + v_2^2}$ for $v = (v_1, v_2) \in \mathbb{R}^2$.

(a) Two vectors $x = (x_1, x_2)$ and $y = (y_1, y_2)$ in $\mathbb{R}^2$ are said to be *orthogonal* if $x_1 y_1 + x_2 y_2 = 0$. Show that for any two orthogonal vectors $u, v \in \mathbb{R}^2$,

$$f(u+v) = f(u) + f(v).$$

(b) Prove that there exists a real constant c such that $f(v) = c\|v\|^2$ for every $v \in \mathbb{R}^2$.

Question 5

15 points

Let $f(t)$ be any polynomial with real coefficients and with degree greater than 1. From the graph of f one can observe that there exists a real number k for which not all roots of $f(t) + k$ are real. (You do not need to prove this analytically; just visualise the graph of f and note that adding k simply shifts the graph vertically — upward if $k > 0$ and downward if $k < 0$.) Use this observation in the problem below.

Let $p(t)$ be a polynomial with the property that $p(t)$ is real whenever t is real, and $p(t)$ is non-real whenever t is non-real. Prove that $p(t)$ must be linear, by following the steps outlined below.

(a) First show that $p(t)$ is real if and only if t is real.

(b) Then prove that the same equivalence holds for $p(t) + k$, for every real number k.

(c) Use part (b) to prove that every root of $p(t) + k$ is real, for every real k.

(d) Finally, conclude that the degree of $p(t)$ cannot exceed 1. Since $p(t)$ cannot be a constant polynomial (why?), it must be linear.

Question 6

15 points

A point $(x, y) \in \mathbb{R}^2$ is said to be a *lattice point* (or a *square lattice point*) if both x and y are integers. We are interested in regular polygons whose vertices are all lattice points.

(a) Prove that there is no regular pentagon with all its vertices at lattice points. Prove the same result for the regular hexagon and for the regular heptagon. In a similar manner, generalise the result to the regular n-gon for every $n > 4$.

(b) What about $n = 3$?

(c) Finally, conclude that the only regular polygons that can be constructed with all vertices at square lattice points are squares themselves.

End of Part B